Chapter 8. Bootstrap Confidence Intervals

Normal, standard, bootstrap-t, percentile and BCa intervals

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Setting

From bootstrap replicates $\hat\theta^{*1},\dots,\hat\theta^{*B}$ (distribution G^*, quantiles Q_{G^*}) we build a $1-\alpha$ interval for θ. The methods differ in what they assume is (approximately) a pivot.

8.1 Normal and basic intervals

Normal-approximation bootstrap interval. If $(\hat\theta-\text{bias}-\theta)/\,\text{se}\,\dot\sim N(0,1)$,

$$\theta \in \hat\theta - \widehat{\text{bias}}_{\text{Boot}} \pm z_{\alpha/2}\widehat{\text{se}}_{\text{Boot}}.$$

It uses only two summaries of G^* and is only as good as the normal approximation.

Basic interval. Assume $\hat\theta-\theta\sim H$ with H free of θ (a pivot) and approximate H by the law H^* of $\hat\theta^*-\hat\theta$, whose quantiles are $Q_{G^*}(p)-\hat\theta$. Since $P(Q_H(\frac{\alpha}{2})\le\hat\theta-\theta\le Q_H(1-\frac{\alpha}{2}))=1-\alpha$,

$$\theta\in\big(2\hat\theta-Q_{G^*}(1-\tfrac{\alpha}{2}),\ 2\hat\theta-Q_{G^*}(\tfrac{\alpha}{2})\big). \tag{8.1}$$

8.2 The bootstrap-t interval

Assume instead that the studentized quantity $t=(\hat\theta-\theta)/\widehat{\text{se}}$ is (approximately) pivotal, and approximate its law by that of

$$t^*=\frac{\hat\theta^*-\hat\theta}{\widehat{\text{se}}^*},$$

where $\widehat{\text{se}}^*$ is the standard error estimate recomputed on each bootstrap sample. With Q_{t^*} the quantiles of the t^{*b},

$$\theta\in\big(\hat\theta-\widehat{\text{se}}\,Q_{t^*}(1-\tfrac{\alpha}{2}),\ \hat\theta-\widehat{\text{se}}\,Q_{t^*}(\tfrac{\alpha}{2})\big). \tag{8.2}$$

If $\widehat{\text{se}}$ has a formula (e.g. $s/\sqrt{n}$ for a mean, or regression standard errors), this is cheap. Otherwise each $\widehat{\text{se}}^*$ needs a second-level bootstrap ($B\times C$ resamples, with $C\approx 25$ usually enough). The method works best on a scale where t is close to pivotal. E.g. for a variance, apply it to $\log\hat\theta$ and exponentiate the endpoints; an increasing transformation preserves coverage.

8.3 Percentile intervals

Percentile interval. $\theta\in(Q_{G^*}(\frac{\alpha}{2}),Q_{G^*}(1-\frac{\alpha}{2}))$.

Proposition 8.1

Suppose there is an increasing φ, not necessarily known, with $\varphi(\hat\theta)-\varphi(\theta)\sim N(0,1)$, and the same holds in the bootstrap world: $\varphi(\hat\theta^*)-\varphi(\hat\theta)\sim N(0,1)$ given the data. Then the percentile interval is exact.

Proof. The exact interval is $\varphi(\theta)\in\varphi(\hat\theta)\pm z_{\alpha/2}$, i.e. $\theta\in(\varphi^{-1}(\varphi(\hat\theta)-z_{\alpha/2}),\varphi^{-1}(\varphi(\hat\theta)+z_{\alpha/2}))$. In the bootstrap world $\varphi(\hat\theta^*)\sim N(\varphi(\hat\theta),1)$, so $\varphi(\hat\theta)\pm z_{\alpha/2}$ are its $\frac{\alpha}{2}$ and $1-\frac{\alpha}{2}$ quantiles. Since φ is increasing, applying φ^{-1} maps them to the quantiles $Q_{G^*}(\frac{\alpha}{2})$, $Q_{G^*}(1-\frac{\alpha}{2})$ of $\hat\theta^*$. The unknown φ never has to be computed. ■

The percentile interval is transformation-respecting and always lies in the parameter space. Its weak point is the assumption that $\varphi(\hat\theta)$ is median-unbiased with constant variance.

8.3.1 Bias-corrected and accelerated (BCa) intervals

Generalize the assumption to

$$\varphi(\hat\theta) - \varphi(\theta) \sim N(-b\,\sigma_\varphi,\ \sigma_\varphi^2), \qquad \sigma_\varphi = 1 + a\,\varphi(\theta), \tag{8.3}$$

with a bias constant b and an acceleration a (the rate at which the standard error changes with the parameter).

Proposition 8.2

Under (8.3) and its bootstrap analogue, the interval $(Q_{G^*}(p_l), Q_{G^*}(p_u))$ is exact, where with $z_l = \Phi^{-1}(\frac{\alpha}{2})$, $z_u = \Phi^{-1}(1-\frac{\alpha}{2})$,

$$p_l = \Phi\Big(b + \frac{b + z_l}{1 - a(b + z_l)}\Big), \qquad p_u = \Phi\Big(b + \frac{b + z_u}{1 - a(b + z_u)}\Big).$$

Proof. Write $\hat\varphi = \varphi(\hat\theta)$, $\phi = \varphi(\theta)$, and assume $1 + a\phi > 0$. Then $Z = \frac{\hat\varphi - \phi}{1+a\phi} + b \sim N(0,1)$. Solving, $\phi = g(Z) = \frac{\hat\varphi - (Z-b)}{1+a(Z-b)}$ with $g'(Z) = -\frac{1+a\hat\varphi}{(1+a(Z-b))^2} < 0$. So $\phi \in [g(-z_l), g(-z_u)] = [L, U]$ iff $Z \in [z_l, z_u]$, which has probability $1-\alpha$; here $L = \frac{\hat\varphi + b + z_l}{1-a(b+z_l)}$ and similarly U. In the bootstrap world, $\varphi(\hat\theta^*) \sim N(\mu_*, \sigma_*^2)$ with $\mu_* = \hat\varphi - b(1+a\hat\varphi)$, $\sigma_* = 1 + a\hat\varphi$. With $w = b + z_l$, $L - \hat\varphi = \frac{w(1+a\hat\varphi)}{1-aw}$, so $P^*(\varphi(\hat\theta^*) \le L) = \Phi(\frac{L-\mu_*}{\sigma_*}) = \Phi(\frac{w}{1-aw} + b) = p_l$. Thus $\varphi^{-1}(L) = Q_{G^*}(p_l)$, and similarly for U. ■

Estimating b and a. Under the model, $P^*(\hat\theta^* \le \hat\theta) = \Phi(b)$, so

$$\hat b = \Phi^{-1}\Big(\frac{\#\{\hat\theta^{*b} \le \hat\theta\}}{B}\Big).$$

For a, use the jackknife: with $\hat\theta_{(i)}$ computed without observation i and $\hat\theta_{(\cdot)}$ their mean,

$$\hat a = \frac{\sum_i (\hat\theta_{(\cdot)} - \hat\theta_{(i)})^3}{6\big[\sum_i (\hat\theta_{(\cdot)} - \hat\theta_{(i)})^2\big]^{3/2}},$$

a skewness measure of the influence of each observation. The jackknife (leave-one-out recomputation; Quenouille 1949, Tukey 1958) was originally proposed for bias and standard-error estimation and is a precursor of the bootstrap. With $a = 0$ the method reduces to the **bias-corrected (BC)** interval, with $p = \Phi(2\hat b + z)$.

8.4 Accuracy and recommendations

For a smooth statistic, the coverage of an interval C satisfies $P(\theta \in C) = 1 - \alpha + O(n^{-k/2})$:

- first-order accurate ($k = 1$): normal theory, normal bootstrap, basic, percentile, BC;
- second-order accurate ($k = 2$): bootstrap-t, BCa.

Recommendation: bootstrap-t when a reliable standard error formula exists (it then needs no second-level resampling); otherwise BCa. Use $B \ge 2000$ for intervals, since tail quantiles are needed. The percentile interval is the simplest default and is often adequate for large samples. For multi-sample or regression resampling, the jackknife estimate of a is less reliable.